Effective Stiffness Tensor for the Perfectly Matched Layer: A Transversely Isotropic Representation

Junkyo Seo

March 2026

1 Form Invariance of the Elastic Wave Equation

The governing equation of elastic waves, the Navier equation, can be written as

$$div\mathbf{C}[grad\mathbf{u}] + \rho\omega^2\mathbf{u} = 0 \tag{1}$$

where $\mathbf{C}$ is the fourth-order elastic stiffness tensor, $\mathbf{u}$ is the displacement vector, ρ is the mass density, and ω is the angular frequency. As the cornerstone of transformation elastodynamics, the form invariance of Eq. (1) under coordinate transformations has been investigated through several approaches. This freedom arises from the fact that the displacement, as a physical vector field, can be represented in different ways under a coordinate map, and the choice of representation is referred to as a *gauge*. Distinct formulations of the transformed equation are therefore obtained according to the choice of gauge.

Among these, two approaches are most commonly discussed. For the coordinate transformation from the virtual domain x^i to the physical domain X^I, let the Jacobian matrix be defined as

$$\lambda^i_I = \frac{\partial x^i}{\partial X^I}.$$

The first approach takes the displacement gauge as λ_i^I, whereas the second approach adopts the identity transformation as the gauge, i.e., δ_i^I. As a consequence of these different gauge choices, the two approaches yield distinct formulations.

In the first approach, with the gauge given by the Jacobian matrix, the Navier equation no longer retains its original form; instead, it transforms into the Willis form, in which the stress depends not only on the displacement but also on the momentum. The corresponding effective material parameters are

$$\begin{aligned} \tilde{C}^{IJKL} &= J^{-1}\Lambda_i^I\Lambda_j^J\Lambda_k^K\Lambda_l^L C^{ijkl} \\ \rho^{IJ} &= J^{-1}\Lambda_i^I\Lambda_j^J\delta^{ij}\rho \end{aligned} \tag{2}$$

where $J = det(\Lambda_i^I)$ is the Jacobian. Here, the density is promoted to an anisotropic tensor, while the stiffness tensor $\tilde{\mathbf{C}}$ retains both major and minor symmetries.

In the second approach, the gauge is taken as the identity. The governing equation in the physical domain again takes the form of the Navier equation, with effective material parameters

$$\begin{aligned} \tilde{C}^{IJKL} &= J^{-1}\Lambda_i^I\delta_j^J\Lambda_k^K\delta_l^L C^{ijkl} \\ \tilde{\rho} &= J^{-1}\rho \end{aligned} \tag{3}$$

Here, the density remains a scalar, multiplied by the inverse Jacobian, but the stiffness tensor $\tilde{\mathbf{C}}$ loses its minor symmetries.[5]

Each approach has its limitations: the first alters the form of the governing equation by introducing the Willis coupling, while the second sacrifices the minor symmetries of the stiffness tensor. Hence, neither yields a fully classical Cauchy elastic representation. From the standpoint of form invariance of the governing equation, however, the second formulation is the more natural choice and is the one adopted in the present work.

In particular, when the Jacobian matrix Λ takes a special form, the transformed stiffness tensor $\tilde{\mathbf{C}}$ admits a natural interpretation as a polar stiffness tensor with a distinctive structure. This is particularly relevant in the context of a perfectly matched layer (PML), in which the complex coordinate stretching is applied independently along each coordinate axis. As a result, the associated Jacobian matrix Λ takes a diagonal form.

In what follows, we restrict attention to a PML applied along the x-direction. The resulting effective medium is interpreted as transversely isotropic with respect to the x-axis, and the transformed stiffness tensor is regarded as that of a transversely isotropic polar solid.

2 Effective PML Medium as a Polar Solid

2.1 Loss of Minor Symmetry under the Identity Gauge

As discussed in the previous section, when the identity transformation is adopted as the displacement gauge, the transformed equation preserves the Navier form, but the transformed stiffness tensor generally loses its minor symmetries. In the present notation, the effective stiffness tensor reads

$$\tilde{C}^{IJKL} = J^{-1}\Lambda_i^I \delta_j^J \Lambda_k^K \delta_l^L C^{ijkl}. \tag{4}$$

Although this tensor preserves the major symmetry that follows from the existence of a quadratic energy density,

$$\tilde{C}^{IJKL} = \tilde{C}^{KLIJ}, \tag{5}$$

it does not generally preserve the minor symmetries

$$\tilde{C}^{IJKL} = \tilde{C}^{JIKL}, \qquad \tilde{C}^{IJKL} = \tilde{C}^{IJLK}. \tag{6}$$

Therefore, the transformed PML stiffness cannot, in general, be interpreted as the stiffness tensor of a classical Cauchy elastic solid.

In a classical Cauchy elastic medium, the strain energy is written in terms of the symmetric strain tensor. Consequently, the stress tensor is symmetric, and the local angular-momentum balance is satisfied without introducing additional body torques. By contrast, $\tilde{\mathbf{C}}$ acts naturally on the full displacement gradient

$$e_{ij} = u_{i,j}, \tag{7}$$

rather than on its symmetric part alone. The effective constitutive law is therefore written as

$$\sigma^{ij} = a^{ijkl} e_{kl}, \tag{8}$$

where a^{ijkl} denotes the effective polar stiffness tensor. Since a^{ijkl} does not necessarily satisfy the minor symmetries, the resulting stress tensor is not necessarily symmetric:

$$\sigma^{ij} \neq \sigma^{ji}. \tag{9}$$

This is the essential reason why the PML-transformed medium should be interpreted as a non-Cauchy, polar-type effective medium.

2.2 Angular-Momentum Interpretation of the Skew Stress

The polar interpretation provides a systematic way to understand the skew part of the stress tensor. Following the framework of isotropic polar elasticity, the equation of linear momentum is retained in the Cauchy form,

$$\sigma^{ij}{}_{,j} + \rho f^i = \rho \dot{v}^i, \tag{10}$$

where σ^{ij} is the force-stress tensor, f^i is the body force per unit mass, and v^i is the particle velocity.

The essential difference from classical elasticity is that the strain-energy density is allowed to depend on the full displacement gradient,

$$\psi = \psi(e_{ij}), \qquad e_{ij} = u_{i,j}. \tag{11}$$

For a linear polar solid, Hooke's law is then obtained by differentiating the energy density with respect to e_{ij}:

$$\sigma^{ij} = \frac{\partial \psi}{\partial e_{ij}} = a^{ijkl} e_{kl}. \tag{12}$$

Because a^{ijkl} is derived from a quadratic energy density, it satisfies the major symmetry

$$a^{ijkl} = a^{klij}. \tag{13}$$

However, since the energy depends on the full displacement gradient, the minor symmetries are not required:

$$a^{ijkl} \neq a^{jikl}, \qquad a^{ijkl} \neq a^{ijlk}. \tag{14}$$

The local angular-momentum balance can be expressed by introducing the body-torque density c^i,

$$c^i = \epsilon^i{}_{jk} \sigma^{jk}, \tag{15}$$

where $\epsilon^i{}_{jk}$ is the Levi-Civita symbol. Substituting the constitutive law (12) into (15) yields

$$c^i = \epsilon^i{}_{jk} a^{jklm} e_{lm}. \tag{16}$$

In the polar interpretation, therefore, the skew part of the stress is not regarded as an inconsistency. Rather, it is interpreted as an induced body-torque density generated by the minor-symmetry-breaking part of the effective stiffness tensor.

This point is essential for the interpretation of the PML effective medium. The transformed PML stiffness does not restore the classical Cauchy condition

$$\sigma^{ij} = \sigma^{ji}. \tag{17}$$

Instead, the nonzero skew stress is interpreted as an effective body torque in the polar sense. The transformed PML medium should therefore be understood as a coordinate-stretched artificial polar medium, rather than as a physically realizable classical elastic material.

2.3 One-Directional PML Stretch and Transverse Isotropy

We now specialize this interpretation to the case in which the PML is applied only in the x^1-direction. In the local principal coordinate system of the stretch, the Jacobian has the diagonal form

$$\Lambda = \mathrm{diag}(a, 1, 1), \tag{18}$$

where $a = a(x^1, \omega)$ denotes the stretch factor in the PML direction.

This one-directional stretching distinguishes the x^1-axis from the transverse plane spanned by x^2 and x^3. Consequently, the transformed stiffness tensor is naturally interpreted as a transversely isotropic polar stiffness tensor with symmetry axis

$$n^i = \delta^i_1. \tag{19}$$

Equivalently, the PML direction defines a preferred axis $\mathbf{n}$, and the plane perpendicular to $\mathbf{n}$ remains isotropic.

This observation motivates the tensor decomposition used in the following section. Since the effective stiffness tensor acts on the full displacement gradient e_{ij}, the decomposition must include both the symmetric strain part and the skew-symmetric rotation part. The appearance of the rotation vector ω^i in the subsequent formulation should therefore not be interpreted as the introduction of an independent microrotation degree of freedom; rather, ω^i is simply the axial-vector representation of the skew-symmetric part of e_{ij}.

The role of the transversely isotropic polar representation is thus twofold. First, it provides a systematic way to parameterize the non-Cauchy stiffness tensor induced by the PML coordinate stretch. Second, it identifies how the loss of minor symmetry activates a coupling between shear deformation and local rotation. This coupling will later appear through the coefficient γ, which measures the shear–rotation interaction generated by the PML-transformed stiffness.

3 Transversely Isotropic Polar Stiffness Tensor Analysis

3.1 Displacement Gradient Decomposition for Transverse Isotropy

We first examine how the stiffness tensor of a transversely isotropic polar solid is represented in a general three-dimensional setting. Previous work by Nassar et al.[4] derived the corresponding representation for an isotropic polar solid. In their formulation, the displacement gradient is decomposed into three rotationally invariant subspaces under full isotropy. These subspaces correspond to the spherical, symmetric-deviatoric, and skew-symmetric components of the gradient, expressed as follows. For the displacement gradient $\mathbf{e}$,

$$\mathbf{e} = \frac{t}{3}\mathbf{I} + \tilde{\mathbf{e}} + \mathbf{W} \tag{20}$$

where $t = tr(\mathbf{e})$, $\mathbf{W} = \frac{\mathbf{e}-\mathbf{e}^{\mathbf{T}}}{2}$, and $\tilde{\mathbf{e}} = \frac{\mathbf{e}+\mathbf{e}^T}{2} - \frac{tr(\mathbf{e})}{3}\mathbf{I}$.

Unlike an isotropic medium, however, a transversely isotropic medium possesses only axial symmetry about a single preferred axis. Direct application of the isotropic decomposition is therefore insufficient: the decomposition must be refined so that each component remains invariant under rotations about the symmetry axis. In other words, the irreducible decomposition must be performed under the restricted symmetry of transverse isotropy.

Under the assumption of axial symmetry about an axis $\mathbf{n}$, the three irreducible components of the isotropic decomposition further split as follows. First, the spherical (volumetric) part decomposes into the axial stretch along $\mathbf{n}$ and the in-plane areal dilation. The skew-symmetric part decomposes into the axial spin about $\mathbf{n}$ and the in-plane (transverse) rotation. Finally, the symmetric deviatoric part decomposes into the in-plane (transverse) deviatoric strain and the out-of-plane (axial) shear strain. These splittings can be summarized as

$$\mathbf{e} = \varepsilon_{nn}(n \otimes n) + \frac{1}{2}\varepsilon_{\perp}(I - n \otimes n) + \mathbf{q} + E_{\perp} + \omega_{\parallel}\mathbf{n} + \omega_{\perp} \tag{21}$$

Denoting by **P** the projection matrix onto the plane perpendicular to the axis **n**, each term is given by

- **Projection tensor:**
$$P = I - n \otimes n$$

- **Axial normal strain:**
$$\varepsilon_{nn} = n \cdot (\varepsilon n)$$

- **In-plane volumetric strain:**
$$\varepsilon_\perp = \mathrm{tr}(P\varepsilon)$$

- **In-plane deviatoric strain:**
$$E_\perp = P\varepsilon P - \frac{\varepsilon_\perp}{2} P$$

- **Out-of-plane shear strain:**
$$q = P\varepsilon n$$

- **Rotation vector:**
$$\omega = \omega_\| n + \omega_\perp$$

Here, the rotation vector $\boldsymbol{\omega}$ is the axial-vector dual of the skew-symmetric part $\mathbf{W}$, defined via the Levi-Civita symbol.

3.2 Derivation of the Transversely Isotropic Polar Stiffness Tensor

We now construct the strain energy ψ as a quadratic form in the irreducible components above, and from it derive the stiffness tensor of a transversely isotropic polar medium. The first task is to identify which combinations of these components yield scalars that are invariant under rotations about the axis **n**. This will be accomplished using Schur's lemma.

3.2.1 Symmetry Restrictions via Schur's Lemma

Let G denote the material symmetry group acting on the deformation tensor space (e.g., the orthogonal group $O(3)$ or the special orthogonal group $SO(3)$). Suppose that this tensor space decomposes into two irreducible representations (irreps) V and W of G. The linear (intertwining) operator $T : V \to W$ is then defined as the operator that maps the tensor state in the irreducible subspace V to the subspace W, thereby producing the cross terms of the strain-energy density.

Since the constitutive equations must respect the material symmetry encoded in G, the linear map T must be equivariant with respect to the group action: that is, for any $g \in G$ and any $x \in V$,

$$T(g \cdot x) = g \cdot T(x) \quad (\forall g \in G,\ \forall x \in V). \tag{22}$$

For a G-equivariant linear map T, Schur's lemma from the theory of group representations yields the following two consequences, depending on the symmetry properties of the irreps.

Lemma 1: Zero Map for Non-isomorphic Irreducible Representations.

If V and W are non-isomorphic irreducible representations possessing distinct symmetry properties, then the only equivariant linear map between them is the zero map. The coupling matrix therefore reduces to $T = 0$, which physically implies that no cross terms in the strain energy can exist between deformation modes of differing symmetry.

Lemma 2: Scalar Multiplicity for Isomorphic Irreducible Representations.

If V and W are isomorphic irreducible representations sharing identical symmetry properties ($V \cong W$), then any equivariant linear map between them must take the form of a scalar multiple of the identity operator I. The coupling matrix is therefore of the form $T = \lambda I$, where λ is a real or complex scalar constant, and λ is identified as an intrinsic elastic stiffness coefficient of the material associated with the corresponding coupling mode.

Applying these results to the irreducible decomposition above, only isomorphic pairs of components — i.e., those sharing the same transformation behavior under rotation about $\mathbf{n}$ — can couple in the strain energy.

3.2.2 Quadratic Form of the Strain Energy Density

Building on this conclusion, each component can be classified by its weight m under rotations of the transverse $SO(2)$ plane.

First, the scalar quantities ε_{nn}, $\varepsilon_\perp$, and $\omega_\parallel$ are invariant under any rotation of the plane and therefore carry weight $m = 0$. The rank-one components $\mathbf{q}$ and $\boldsymbol{\omega}_\perp$ rotate by the same angle θ under a transverse rotation by θ, and so carry weight $m = 1$. The rank-two component $E_\perp$, in contrast, rotates by 2θ and carries weight $m = 2$. Components of the same weight thus share identical transformation properties under transverse rotations.

Equality of weight alone, however, does not guarantee a coupling in the strain energy: the components must also share the same parity under reflections. The rotation components $\omega_\parallel$ and $\boldsymbol{\omega}_\perp$ are axial vectors, and hence are invariant under reflection, whereas the strain components — the axial and in-plane normal strains, the in-plane deviatoric strain, and the out-of-plane shear strain — change sign under reflection. Under the assumption of centrosymmetry, coupling between these two classes is therefore forbidden.

In summary, only components of equal weight that also share parity under reflections can couple in the strain energy. The resulting quadratic form reads

$$\begin{aligned}\psi = \frac{1}{2}A\varepsilon_{nn}^2 + B\varepsilon_{nn}\varepsilon_\perp + \frac{1}{2}C\varepsilon_\perp^2 + DE_\perp : E_\perp + 2L\mathbf{q}\cdot\mathbf{q} \\ + \frac{1}{2}\alpha_\parallel\omega_\parallel^2 + \frac{1}{2}\alpha_\perp\omega_\perp\cdot\omega_\perp + \gamma\omega_\perp\cdot(n\times q)\end{aligned} \tag{23}$$

The last term deserves particular attention. Although $\mathbf{q}$ and $\boldsymbol{\omega}_\perp$ have opposite parities — $\mathbf{q}$ is a polar vector while $\boldsymbol{\omega}_\perp$ is an axial vector — the cross product $\mathbf{n}\times\mathbf{q}$ converts $\mathbf{q}$ into an axial vector, restoring matching parity and thereby allowing the coupling. No analogous mechanism exists among the scalar (weight-zero) components, so no analogous cross-coupling among scalars arises.

3.2.3 Explicit Representation of the Stiffness Tensor

We now derive the stiffness tensor from the quadratic strain-energy density obtained above. Under the assumption of linear elasticity, the stiffness tensor is obtained from the energy density via

$$\frac{\partial^2\psi}{\partial e_{ij}\partial e_{kl}} = a^{ijkl}. \tag{24}$$

To apply this relation, we first rewrite each contribution to ψ in index notation, then differentiate with respect to the displacement gradient e_{ij}.

The strain components derived in Section 3.1 can be expressed in index notation as follows.

- **Projection tensor:**
$$P^{ij} = \delta^{ij} - n^i n^j$$

- **Axial normal strain:**
$$\varepsilon_{nn} = n^i e_{ij} n^j$$

- **In-plane volumetric strain:**
$$\varepsilon_{\perp} = P^{ij} e_{ij}$$

- **In-plane deviatoric strain:**
$$E_{\perp}^{ij} = P^{im}\varepsilon_{mn}P^{nj} - \frac{1}{2}(P^{mn}\varepsilon_{mn})P^{ij}$$

- **Out-of-plane shear strain:**
$$q^i = P^{im}\varepsilon_{mn}n^n$$

- **Rotation vector:**
$$\omega^m = \frac{1}{2}\epsilon^{mij}e_{ij}$$

Substituting these expressions into each energy contribution and rearranging into the canonical form $\psi = \frac{1}{2}a^{ijkl}e_{ij}e_{kl}$ yields the following representations.

- **Term 1:** $\frac{1}{2}A\varepsilon_{nn}^2$
$$\frac{1}{2}A\varepsilon_{nn}^2 = \frac{1}{2}A(n^i n^j e_{ij})(n^k n^l e_{kl}) = \frac{1}{2}(An^i n^j n^k n^l)e_{ij}e_{kl}$$

- **Term 2:** $B\varepsilon_{nn}\varepsilon_{\perp}$
$$\begin{aligned} B\varepsilon_{nn}\varepsilon_{\perp} &= B(n^i n^j e_{ij})(P^{kl}e_{kl}) = Bn^i n^j P^{kl} e_{ij}e_{kl} \\ &= \frac{1}{2}B(n^i n^j P^{kl} + P^{ij} n^k n^l)e_{ij}e_{kl} \end{aligned}$$

- **Term 3:** $\frac{1}{2}C\varepsilon_{\perp}^2$
$$\frac{1}{2}C\varepsilon_{\perp}^2 = \frac{1}{2}C(P^{ij}e_{ij})(P^{kl}e_{kl}) = \frac{1}{2}CP^{ij}P^{kl}e_{ij}e_{kl}$$

- **Term 4:** $DE_{\perp} : E_{\perp}$
$$\begin{aligned} D\, E_{\perp}^{ij}E_{\perp}^{ij} &= D\big(P^{im}\varepsilon_{mn}P^{nj} - \tfrac{1}{2}P^{mn}\varepsilon_{mn}P^{ij}\big)\big(P^{ip}\varepsilon_{pq}P^{qj} - \tfrac{1}{2}P^{pq}\varepsilon_{pq}P^{ij}\big) \\ &= D\Big[\, P^{mp}P^{nq}\varepsilon_{mn}\varepsilon_{pq} - (P^{mn}\varepsilon_{mn})(P^{pq}\varepsilon_{pq}) + \tfrac{1}{2}(P^{mn}\varepsilon_{mn})(P^{pq}\varepsilon_{pq})\,\Big] \\ &= D\Big(P^{ik}P^{jl} - \tfrac{1}{2}P^{ij}P^{kl}\Big)\varepsilon_{ij}\varepsilon_{kl} \\ &= \tfrac{D}{2}\Big(P^{ik}P^{jl} + P^{il}P^{jk} - P^{ij}P^{kl}\Big)e_{ij}e_{kl} \end{aligned}$$

- **Term 5:** $2L\mathbf{q}\cdot\mathbf{q}$

$$
\begin{aligned}
2Lq^m q^m &= 2L(P^{mi}\varepsilon_{ij}n^j)(P^{mk}\varepsilon_{kl}n^l) = \frac{L}{2}(P^{ik}n^jn^l)(e_{ij}+e_{ji})(e_{kl}+e_{lk}) \\
&= \frac{1}{2}L(P^{ik}n^jn^l + P^{il}n^jn^k + P^{jk}n^in^l + P^{jl}n^in^k)e_{ij}e_{kl}
\end{aligned}
$$

- **Term 6:** $\frac{1}{2}\alpha_\parallel\omega_\parallel^2$

$$
\frac{1}{2}\alpha_\parallel\omega_\parallel^2 = \frac{1}{2}\alpha_\parallel(\omega^m n_m)(\omega^n n_n) = \frac{1}{2}(\frac{1}{4}\alpha_\parallel n_m n_n \epsilon^{mij}\epsilon^{nkl})e_{ij}e_{kl}
$$

- **Term 7:** $\frac{1}{2}\alpha_\perp\omega_\perp\cdot\omega_\perp$

$$
\begin{aligned}
\frac{1}{2}\alpha_\perp\omega_\perp^i\cdot\omega_\perp^i &= \frac{1}{2}(\alpha_\perp(P_m^i\omega^m)(P_n^i\omega^n)) = \frac{1}{2}(\alpha_\perp P_{mn}(\omega^m\omega^n)) \\
&= \frac{1}{2}(\frac{1}{4}\alpha_\perp P_{mn}\epsilon^{mij}\epsilon^{nkl})e_{ij}e_{kl}
\end{aligned}
$$

- **Term 8:** $\gamma\omega_\perp\cdot(n\times q)$

$$
\begin{aligned}
\gamma\,\omega_\perp^i\cdot(n\times q)_i &= \gamma\,\omega^i(n\times q)_i \\
&= \gamma\big(\tfrac{1}{2}\epsilon^{kab}\omega_{ab}\big)\big(\epsilon_{kij}n^iq^j\big) \\
&= \tfrac{\gamma}{2}\big(\delta_i^a\delta_j^b - \delta_j^a\delta_i^b\big)\omega_{ab}n^iq^j \\
&= \tfrac{\gamma}{2}\big(\omega_{ij}-\omega_{ji}\big)n^iq^j = \gamma\,\omega_{ij}n^iq^j \\
&= \gamma\,\omega_{ij}n^i\big(\delta^{jc}-n^jn^c\big)\varepsilon_{cd}n^d = \gamma\,\omega_{ij}\varepsilon_{jd}n^in^d \\
&= \tfrac{\gamma}{4}\big(e_{ij}-e_{ji}\big)\big(e_{jd}+e_{dj}\big)n^in^d \\
&= \tfrac{\gamma}{4}\big[\,e_{ij}e_{jd}+e_{ij}e_{dj}-e_{ji}e_{jd}-e_{ji}e_{dj}\,\big]n^in^d \\
&= \tfrac{\gamma}{4}\big[\,\delta^{jk}n^in^l+\delta^{jl}n^in^k-\delta^{ik}n^jn^l-\delta^{il}n^jn^k\,\big]e_{ij}e_{kl} \\
&= \tfrac{1}{2}\Big[-\tfrac{\gamma}{2}\big(\delta^{ik}n^jn^l-\delta^{jl}n^in^k\big)\Big]e_{ij}e_{kl}.
\end{aligned}
$$

Combining all contributions, the transversely isotropic polar stiffness tensor a^{ijkl} takes the form

$$
\begin{aligned}
a^{ijkl} &= An^in^jn^kn^l \\
&+ B(n^in^jP^{kl}+P^{ij}n^kn^l) \\
&+ C(P^{ij}P^{kl}) \\
&+ D(P^{ik}P^{jl}+P^{il}P^{jk}-P^{ij}P^{kl}) \\
&+ L(P^{ik}n^jn^l+P^{il}n^jn^k+P^{jk}n^in^l+P^{jl}n^in^k) \\
&+ \frac{1}{4}(\alpha_\perp P_{mn}+\alpha_\parallel n_mn_n)\epsilon^{mij}\epsilon^{nkl} \\
&+ \frac{\gamma}{2}(\delta^{jl}n^in^k-\delta^{ik}n^jn^l)
\end{aligned}
\tag{25}
$$

Setting the symmetry axis $\mathbf{n}$ along $\mathbf{e}_1$, i.e., the x-axis, the components are

Normal block

$$\begin{aligned} a^{1111} &= A, \\ a^{1122} = a^{1133} = a^{2211} = a^{3311} &= B, \\ a^{2222} = a^{3333} &= C + D, \\ a^{2233} = a^{3322} &= C - D. \end{aligned}$$

$(2,3)$ **shear pair**

$$\begin{aligned} a^{2323} = a^{3232} &= D + \frac{\alpha_\parallel}{4}, \\ a^{2332} = a^{3223} &= D - \frac{\alpha_\parallel}{4}. \end{aligned}$$

$(1,2)$ **shear pair**

$$\begin{aligned} a^{1212} &= L + \frac{\alpha_\perp}{4} + \frac{\gamma}{2}, \\ a^{1221} = a^{2112} &= L - \frac{\alpha_\perp}{4}, \\ a^{2121} &= L + \frac{\alpha_\perp}{4} - \frac{\gamma}{2}. \end{aligned}$$

$(1,3)$ **shear pair**

$$\begin{aligned} a^{1313} &= L + \frac{\alpha_\perp}{4} + \frac{\gamma}{2}, \\ a^{1331} = a^{3113} &= L - \frac{\alpha_\perp}{4}, \\ a^{3131} &= L + \frac{\alpha_\perp}{4} - \frac{\gamma}{2}. \end{aligned}$$

3.3 Interpretation of the PML as an Effective Medium

Consider a PML applied in the $+x$-direction. From Eq. (3) in Section 1, the stiffness tensor of the PML region can be obtained by treating the transformed domain as an effective medium. Because the PML is applied only along the $+x$-axis, the Jacobian matrix takes the form $\Lambda = \mathrm{diag}(a, 1, 1)$.

Following Chang et al. [1], applying the polar decomposition to Λ and using the eigenvalues of the resulting symmetric tensor $\mathbf{V}$, the transformed stiffness tensor admits the form

$$\tilde{C}^{IJKL} = \frac{1}{J}\lambda_I \lambda_K C^{IJKL} \tag{26}$$

where each capital index in the subscript takes the same numerical value as its corresponding lower index, and $J = \prod_{I=1}^{n} \lambda_I$.

Restricting attention to the components relevant to the present discussion, we obtain

$$\begin{gathered}
\tilde{C}^{1111} = a(\lambda + 2\mu), \\
\tilde{C}^{1122} = \tilde{C}^{1133} = \tilde{C}^{2211} = \tilde{C}^{3311} = \lambda, \\
\tilde{C}^{2222} = \tilde{C}^{3333} = \frac{\lambda + 2\mu}{a}, \\
\tilde{C}^{2233} = \tilde{C}^{3322} = \frac{\lambda}{a}.
\end{gathered}$$

$$\begin{gathered}
\tilde{C}^{1212} = a\mu, \qquad \tilde{C}^{1221} = \tilde{C}^{2112} = \mu, \qquad \tilde{C}^{2121} = \frac{\mu}{a}, \\
\tilde{C}^{1313} = a\mu, \qquad \tilde{C}^{1331} = \tilde{C}^{3113} = \mu, \qquad \tilde{C}^{3131} = \frac{\mu}{a}, \\
\tilde{C}^{2323} = \tilde{C}^{2332} = \tilde{C}^{3223} = \tilde{C}^{3232} = \frac{\mu}{a}.
\end{gathered}$$

Matching these components against the parameters of the transversely isotropic polar stiffness tensor derived above yields

$$\boxed{\begin{aligned}
&A = a(\lambda + 2\mu), \\
&B = \lambda, \\
&C = \frac{\lambda + \mu}{a}, \\
&D = \frac{\mu}{a}, \\
&L = \frac{\mu}{4}\left(a + 2 + \frac{1}{a}\right) = \mu\frac{(a+1)^2}{4a}, \\
&\alpha_{\parallel} = 0, \\
&\alpha_{\perp} = \mu\left(a + \frac{1}{a} - 2\right) = \mu\frac{(a-1)^2}{a}, \\
&\gamma = \mu\left(a - \frac{1}{a}\right).
\end{aligned}}$$

3.4 Body-Torque Interpretation of the Shear–Rotation Coupling

The coefficient γ can be interpreted further through the body-torque density introduced in Section 2. For a polar medium, the skew part of the stress is associated with the body-torque density

$$c^i = \epsilon^i{}_{jk}\sigma^{jk}. \tag{27}$$

Hence, a symmetric stress tensor implies $c^i = 0$, while a nonzero skew stress generated by the minor-symmetry-breaking stiffness appears as an induced body torque.

For the present transversely isotropic representation, let the symmetry axis be aligned with the x^1-axis,

$$n^i = \delta^i_1.$$

Then the shear–rotation part of the strain energy is

$$\psi_{\mathrm{sh-rot}} = 2L(\varepsilon_{12}^2 + \varepsilon_{13}^2) + \frac{1}{2}\alpha_\perp(\omega_2^2 + \omega_3^2) + \frac{1}{2}\alpha_\parallel\omega_1^2 + \gamma(\varepsilon_{12}\omega_3 - \varepsilon_{13}\omega_2). \tag{28}$$

Using

$$\sigma^{ij} = \frac{\partial\psi}{\partial e_{ij}},$$

the body-torque components are obtained as

$$c^1 = \sigma^{23} - \sigma^{32} = \alpha_\parallel\omega_1, \tag{29}$$

$$c^2 = \sigma^{31} - \sigma^{13} = \alpha_\perp\omega_2 - \gamma\varepsilon_{13}, \tag{30}$$

$$c^3 = \sigma^{12} - \sigma^{21} = \alpha_\perp\omega_3 + \gamma\varepsilon_{12}. \tag{31}$$

Equivalently, in vector form,

$$c^i = \alpha_\parallel\omega_\parallel n^i + \alpha_\perp\omega_\perp^i + \gamma\epsilon^i{}_{jk}n^j q^k. \tag{32}$$

The coefficient γ therefore does not merely represent an algebraic coupling between the out-of-plane shear strain and the transverse rotation; it also measures the part of the induced body torque generated by the minor-symmetry-breaking PML stiffness. In particular, when $a \neq 1$, the transformed stiffness tensor is generally non-Cauchy, and the associated skew stress can be interpreted as an effective body torque in the polar sense.

It should be emphasized that this interpretation does not introduce an independent microrotation field. The rotation vector ω^i is still determined by the skew-symmetric part of the displacement gradient e_{ij}. The body-torque expression above is therefore a diagnostic representation of the PML-induced non-Cauchy stiffness, rather than a full micropolar balance law with independent rotational degrees of freedom.

3.5 Strain Energy Positivity and Stability Analysis

Specializing the strain-energy density derived earlier to the case $\mathbf{n} = \mathbf{e}_1$, we obtain

$$\begin{aligned}\psi = \frac{1}{2}A\varepsilon_{11}^2 + B\varepsilon_{11}\varepsilon_\perp + \frac{1}{2}C\varepsilon_\perp^2 + DE_\perp : E_\perp + 2L(\varepsilon_{12}^2 + \varepsilon_{13}^2)\\ + \frac{1}{2}\alpha_\perp(\omega_2^2 + \omega_3^2) + \frac{1}{2}\alpha_\parallel\omega_1^2 + \gamma(\varepsilon_{12}\omega_3 - \varepsilon_{13}\omega_2).\end{aligned} \tag{33}$$

By construction, the only couplings allowed by symmetry occur within the same block; modes belonging to different blocks remain decoupled. Consequently, the total strain energy decomposes into four independent additive blocks, each of which can be analyzed separately:

$$\psi_{\mathrm{block}} = \begin{cases} \text{(i) Normal block} & \frac{1}{2}\begin{bmatrix}\varepsilon_{11} & \varepsilon_\perp\end{bmatrix}\begin{bmatrix}A & B\\ B & C\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_\perp\end{bmatrix} \\ \text{(ii) Transverse deviatoric block} & DE_\perp : E_\perp \\ \text{(iii) Axial rotation block} & \frac{1}{2}\alpha_\parallel\omega_1^2 \\ \text{(iv) Shear-rotation block} & \frac{1}{2}\begin{bmatrix}\varepsilon & \omega\end{bmatrix}\begin{bmatrix}4L & \pm\gamma\\ \pm\gamma & \alpha_\perp\end{bmatrix}\begin{bmatrix}\varepsilon\\ \omega\end{bmatrix} \end{cases} \tag{34}$$

Since the PML stretch factor is complex, we now investigate the conditions under which the Hermitian part of each block matrix is positive semidefinite. Define

$$a = \alpha + i\frac{\beta}{\omega}.$$

Substituting this expression into the parameters derived above and taking real parts yields

$$\Re A = (\lambda + 2\mu)\alpha$$

$$\Re B = \lambda$$

$$\Re C = (\lambda + \mu)\frac{\alpha}{|a|^2}$$

$$\Re D = \mu\frac{\alpha}{|a|^2}$$

$$\Re L = \frac{\mu}{4}\left(\alpha + 2 + \frac{x}{|a|^2}\right)$$

$$\Re \alpha_{\parallel} = 0$$

$$\Re \alpha_{\perp} = \mu\left(\alpha + \frac{\alpha}{|a|^2} - 2\right)$$

$$\Re \gamma = \mu\left(\alpha - \frac{\alpha}{|a|^2}\right).$$

We now derive the conditions under which the Hermitian part of each block is positive semidefinite.

(i) Normal block.

$$\frac{1}{2}\begin{bmatrix}\varepsilon_{11} & \varepsilon_{\perp}\end{bmatrix}\begin{bmatrix}A & B\\ B & C\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{\perp}\end{bmatrix}$$

For complex coefficients, the Hermitian part of the block is

$$H_N = \begin{bmatrix}\Re A & B\\ B & \Re C\end{bmatrix},$$

so the positive semidefiniteness conditions read

$$\Re A \geq 0, \qquad \Re C \geq 0, \qquad (\Re A)(\Re C) - B^2 \geq 0.$$

Substitution gives

$$(\lambda + 2\mu)\alpha \geq 0$$

$$(\lambda + \mu)\frac{\alpha}{|a|^2} \geq 0$$

$$(\lambda + 2\mu)(\lambda + \mu)\frac{\alpha^2}{|a|^2} - \lambda^2 \geq 0.$$

Rearranging the last inequality yields

$$\mu(3\lambda + 2\mu)\alpha^2 - \lambda^2 \frac{\beta^2}{\omega^2} \geq 0.$$

The conditions for the normal block are therefore

$$\boxed{\alpha \geq 0, \qquad \mu(3\lambda + 2\mu)\alpha^2 \geq \lambda^2 \frac{\beta^2}{\omega^2}.} \tag{35}$$

(ii) Transverse deviatoric block.

For this block, only D enters, and the condition is

$$\Re D \geq 0,$$

that is,

$$\mu \frac{\alpha}{|a|^2} \geq 0.$$

If $\mu > 0$ in the background medium, this reduces to

$$\boxed{\alpha \geq 0.} \tag{36}$$

(iii) Axial rotation block.

The block reads

$$\frac{1}{2}\alpha_{\|}\omega_1^2,$$

and in the present case

$$\alpha_{\|} = 0.$$

Hence

$$\boxed{\alpha_{\|} = 0.} \tag{37}$$

(iv) Shear-rotation block.

For example, the (12)-ω_3 block reads

$$2L\varepsilon_{12}^2 + \gamma\varepsilon_{12}\omega_3 + \frac{1}{2}\alpha_\perp\omega_3^2,$$

and its Hermitian part is

$$H_S = \begin{bmatrix} 4\Re L & \Re\gamma \\ \Re\gamma & \Re\alpha_\perp \end{bmatrix}.$$

The positive semidefiniteness conditions are

$$4\Re L \geq 0, \qquad \Re\alpha_\perp \geq 0, \qquad (4\Re L)(\Re\alpha_\perp) - (\Re\gamma)^2 \geq 0.$$

Substituting the explicit forms of the real parts,

$$4\Re L = \mu\left(\alpha + 2 + \frac{\alpha}{|a|^2}\right),$$

$$\Re\alpha_\perp = \mu\left(\alpha + \frac{\alpha}{|a|^2} - 2\right),$$

$$\Re\gamma = \mu\left(\alpha - \frac{\alpha}{|a|^2}\right),$$

the determinant becomes

$$\begin{aligned}&(4\Re L)(\Re\alpha_\perp) - (\Re\gamma)^2\\ &= \mu^2\left(\alpha + 2 + \frac{\alpha}{|a|^2}\right)\left(\alpha + \frac{\alpha}{|a|^2} - 2\right) - \mu^2\left(\alpha - \frac{\alpha}{|a|^2}\right)^2.\end{aligned}$$

Expansion yields the following compact expression:

$$\boxed{(4\Re L)(\Re\alpha_\perp) - (\Re\gamma)^2 = -\frac{4\mu^2\beta^2}{\omega^2|a|^2} = -\frac{4\mu^2\beta^2}{\omega^2\alpha^2 + \beta^2} \le 0} \tag{38}$$

From Eq. (38), the determinant of the Hermitian part of the shear–rotation block is non-positive in general, and strictly negative whenever the stretch factor a is genuinely complex (i.e., $\beta \neq 0$). In this regime the Hermitian positive-semidefiniteness condition fails, and the quadratic form associated with the coupling between the out-of-plane shear strain $\mathbf{q}$ and the transverse rotation $\boldsymbol{\omega}_\perp$ is not bounded below. The PML-induced effective medium therefore admits local energy divergence within this deformation sector.

This conclusion parallels the failure mechanism identified by Loh et al. [3] for PMLs applied to backward-wave structures. In their analysis, the PML is interpreted not as a purely isotropic lossy absorber, but as an anisotropic absorbing medium whose principal directions may include a direction of effective gain. For backward-wave modes, the field component aligned with this gain direction dominates the modal energy budget, so that the PML produces net amplification rather than attenuation.

In the present elastodynamic setting, the non-positivity of the shear–rotation determinant furnishes a local analogue of this mechanism. Whenever the stretch factor is genuinely complex, the coupling between $\mathbf{q}$ and $\boldsymbol{\omega}_\perp$ generates an indefinite energy sector — a quadratic form whose Hermitian part lacks sign-definiteness. The shear–rotation block may therefore be regarded as the local elastodynamic counterpart of the gain channel reported by Loh et al.: it isolates the specific deformation sector in which the effective PML medium ceases to be locally passive. While this observation does not, in itself, establish global time-domain instability, it identifies the specific block through which modal energy amplification can enter the elastodynamic PML formulation.

4 Analysis of the Stability Condition

4.1 Connection to Previous Studies

Previous studies [2] have empirically observed that long-time stability is ensured when the real part of the stretch factor exceeds its imaginary part:

$$\left|\mathrm{Re}\{\hat{L}\}\right| > C\left|\mathrm{Im}\{\hat{L}\}\right|, \qquad 0 < C \le 1, \tag{39}$$

where $\hat{L}$ denotes the stretch factor of the perfectly matched layer in the absorbing region.

Physically, this condition requires that the wave can propagate sufficiently within the PML region before being attenuated, in order to prevent numerical divergence. On this basis, it is naturally associated with the normal block, which governs the volumetric response along the PML axis.

To relate the effective-medium positivity conditions derived above to the parameter-dependent stability criterion of the PML formulation, we now specialize the complex stretch factor to the following one-dimensional PML setting. The computational domain is decomposed into the interior domain and the PML region as

$$(0, L) = (0, L_{\mathrm{ID}}) \cup (L_{\mathrm{ID}}, L), \qquad L_{\mathrm{PML}} = L - L_{\mathrm{ID}}.$$

The physical coordinate x is replaced by the complex stretched coordinate $X(x)$, defined by

$$x \mapsto X(x) = \int_0^x \lambda(s, \omega)\, ds,$$

where the stretch function reads

$$\lambda(s, \omega) = a(s) + \frac{b(s)}{i\omega + \omega_0}.$$

Here, ω is the angular frequency and $\omega_0 \geq 0$ is the complex-frequency-shift parameter. The real-stretching profile $a(s)$ and the damping profile $b(s)$ are prescribed as

$$a(s) = \begin{cases} 1, & s \in (0, L_{\mathrm{ID}}), \\ 1 + \alpha \left(\dfrac{s - L_{\mathrm{ID}}}{L_{\mathrm{PML}}} \right)^m, & s \in (L_{\mathrm{ID}}, L), \end{cases}$$

and

$$b(s) = \begin{cases} 0, & s \in (0, L_{\mathrm{ID}}), \\ \beta \left(\dfrac{s - L_{\mathrm{ID}}}{L_{\mathrm{PML}}} \right)^m, & s \in (L_{\mathrm{ID}}, L). \end{cases}$$

Thus, no coordinate stretching is applied in the interior domain, while a polynomial stretching profile of order m is introduced inside the PML region. Within the PML layer, the stretch function can be rewritten as

$$\lambda(s, \omega) = 1 + \left[\alpha + \frac{\beta}{i\omega + \omega_0} \right] \left(\frac{s - L_{\mathrm{ID}}}{L_{\mathrm{PML}}} \right)^m, \qquad s \in (L_{\mathrm{ID}}, L).$$

First, consider the classical PML case in which $\omega_0 = 0$. Under the above setting, Eq. (35) can be rewritten as

$$\mu(3\lambda + 2\mu)(1 + \alpha\xi^m)^2 \geq \lambda^2 \left(\frac{\beta\xi^m}{\omega} \right)^2 \tag{40}$$

where $\xi = \frac{s - L_{ID}}{L_{PML}} \in (0, 1)$.

Assuming positivity of the relevant coefficients, taking the square root of both sides yields

$$1 + \alpha\xi^m \geq k \frac{\beta\xi^m}{\omega} \tag{41}$$

where $k = \sqrt{\frac{\lambda^2}{\mu(3\lambda + 2\mu)}}$.

For $\xi > 0$, this condition is equivalent to

$$\alpha \geq k\frac{\beta}{\omega} - \xi^{-m}. \tag{42}$$

Since ξ^{-m} is minimized at $\xi = 1$ for $m \geq 0$, the right-hand side of Eq. (42) is maximized at the outer edge of the PML layer. The most restrictive pointwise condition is therefore obtained at $\xi = 1$:

$$\alpha \geq k\frac{\beta}{\omega} - 1. \tag{43}$$

Next, consider the CFS-PML case with $\omega_0 \neq 0$. Applying the same setting, the inequality for the normal block becomes

$$1 + \xi^m \left(\alpha + \frac{\beta\omega_0}{\omega^2 + \omega_0^2} \right) \geq k\xi^m \frac{\beta\omega}{\omega^2 + \omega_0^2}. \tag{44}$$

This can be rewritten as

$$\alpha \geq \frac{\beta(k\omega - \omega_0)}{\omega^2 + \omega_0^2} - \xi^{-m}. \tag{45}$$

Again, setting $\xi = 1$ yields the most restrictive pointwise condition:

$$\alpha \geq \frac{\beta(k\omega - \omega_0)}{\omega^2 + \omega_0^2} - 1. \tag{46}$$

This result has a noteworthy implication. The empirical long-time stability condition proposed in previous studies (with $C = 1$) reads

$$\begin{cases} \alpha \geq \dfrac{\beta}{\omega} - 1, & \text{for the classical PML,} \\ \alpha \geq \dfrac{\beta(\omega - \omega_0)}{\omega^2 + \omega_0^2} - 1, & \text{for the CFS-PML.} \end{cases} \tag{47}$$

The empirical condition closely matches the pointwise condition derived from the positive semidefiniteness of the normal block; the only difference is the appearance of the material-dependent factor k. Both stability criteria therefore lead to essentially the same conclusion.

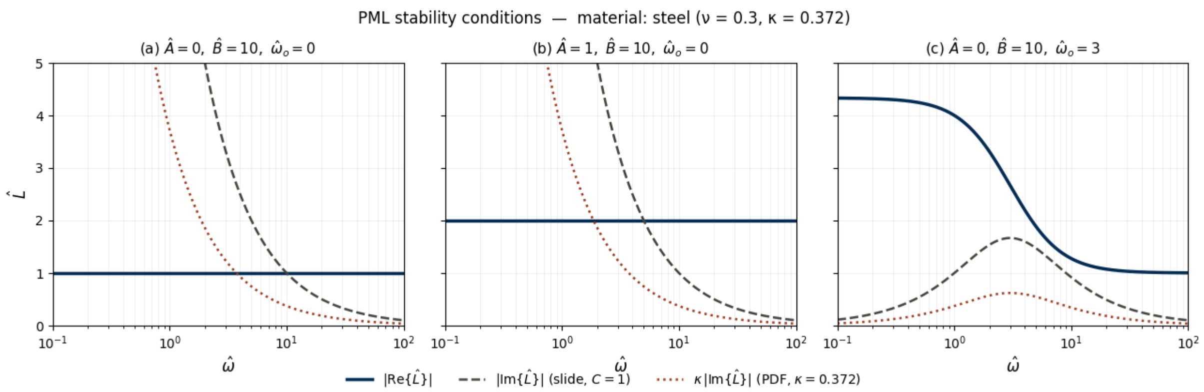

Figure 1: Comparison of PML stability conditions for steel with $\nu = 0.3$ and $k = 0.372$. The solid blue curve represents $|\operatorname{Re}\{\hat{L}\}|$, the dashed gray curve represents $|\operatorname{Im}\{\hat{L}\}|$ from the empirical condition with $C = 1$, and the dotted red curve represents $k\,|\operatorname{Im}\{\hat{L}\}|$ from the pointwise, material-dependent condition.

As Fig. 1 shows, in the classical PML the imaginary part of $\hat{L}$ diverges as $\omega \to 0$, so there always exists a frequency at which the stability condition fails. In contrast, the CFS-PML with $\omega_0 \neq 0$ prevents this divergence. Moreover, as illustrated in the right panel of the figure, an appropriate choice of ω_0 ensures that the real part of $\hat{L}$ exceeds k times its imaginary part for all ω. Hence, with a suitable ω_0, the CFS-PML can guarantee stability across the entire frequency range, whereas the classical PML cannot, with instability typically arising in the low-frequency regime.

A second point worth emphasizing concerns the factor k, which encodes the material dependence of the domain. Expressing k in terms of Poisson's ratio yields

$$k(\nu) = \frac{2\nu}{\sqrt{2(1+\nu)(1-2\nu)}}. \tag{48}$$

Solving $k(\nu) = 1$ gives $\nu \approx 0.387$, a value that lies within the typical range of Poisson's ratios for many metals and for concrete.

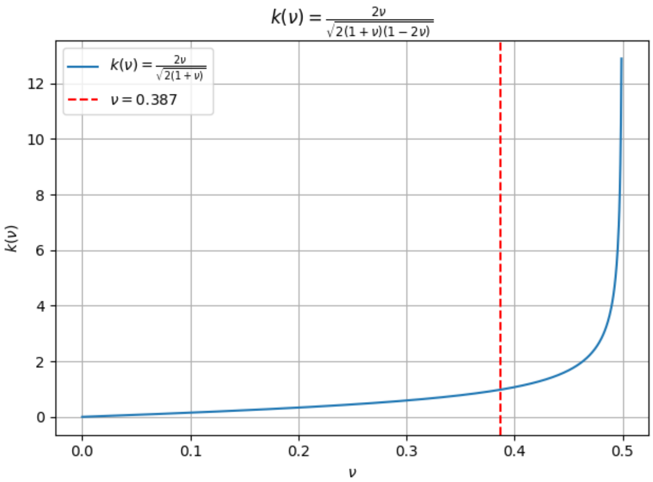

Figure 2: Variation of $k(\nu)$ with Poisson's ratio ν.

As Fig. 2 shows, $k < 1$ whenever $\nu < 0.387$. In this regime, the empirical condition in Eq. (47) is more restrictive than the material-dependent conditions in Eqs. (42) and (45), which explains why prior simulations using the empirical criterion typically observed stable behavior.

The empirical condition, however, may also produce incorrect stability assessments in some regimes — either by being overly restrictive or by being insufficiently restrictive.

As shown in Fig. 3, when the Poisson's ratio is below the critical value $\nu \approx 0.387$ (where $k = 1$), the empirical inequality is overly restrictive, so it may falsely predict numerical divergence in cases where the simulation is in fact stable. Conversely, when $\nu > 0.387$, the empirical inequality may erroneously predict stability while the simulation in fact diverges. In summary, for engineering materials with $\nu < 0.387$ — which covers most metals and concrete — the empirical criterion serves as a conservative practical condition with a built-in safety margin against numerical divergence. For materials with $\nu > 0.387$, however, it can produce misleading stability assessments. The inequality derived in the present work must therefore be validated through numerical simulations, with particular attention to the regimes identified in Fig. 3.

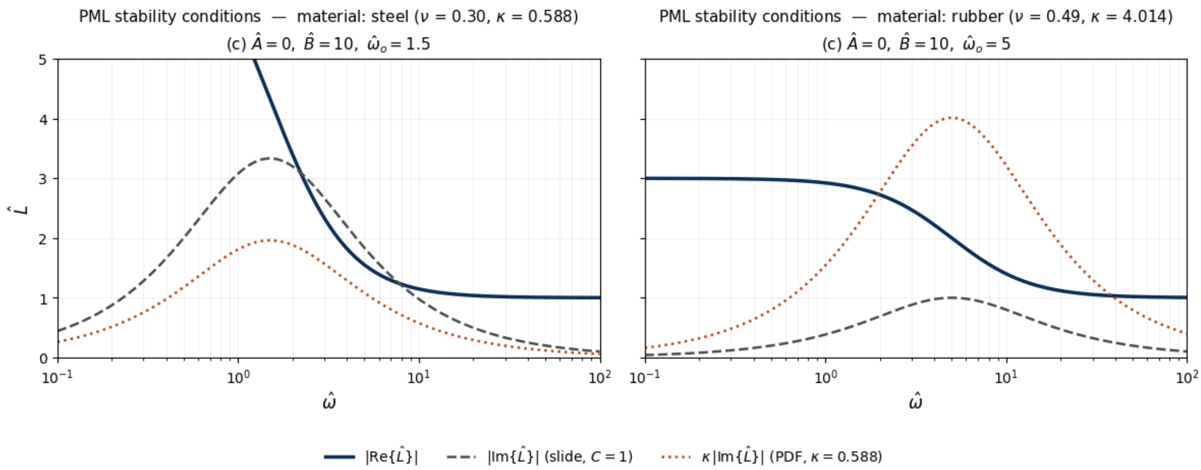

Figure 3: Comparison of the empirical and material-dependent PML stability conditions for two representative materials. In both cases, $\hat{A} = 0$ and $\hat{B} = 10$. The left panel corresponds to steel with $\nu = 0.30$, $k = 0.588$, and $\hat{\omega}_0 = 1.5$; the right panel corresponds to rubber with $\nu = 0.49$, $k = 4.014$, and $\hat{\omega}_0 = 5$. For steel ($k < 1$), the empirical condition is more restrictive, whereas for rubber ($k > 1$), the empirical condition may become insufficiently restrictive.

References

[1] Zheng Chang, Dengke Guo, Xi-Qiao Feng, and Gengkai Hu. A facile method to realize perfectly matched layers for elastic waves. *Wave Motion*, 51(7):1170–1178, 2014.

[2] Heedong Goh, Loukas F. Kallivokas, and Stijn François. On the stability of perfectly matched layers in elastodynamics. Presentation slides, May 2025.

[3] Po-Ru Loh, Ardavan F. Oskooi, Mihai Ibanescu, Maksim Skorobogatiy, and Steven G. Johnson. Fundamental relation between phase and group velocity, and application to the failure of perfectly matched layers in backward-wave structures. *Phys. Rev. E*, 79:065601(R), Jun 2009.

[4] H. Nassar, Y.Y. Chen, and G.L. Huang. Isotropic polar solids for conformal transformation elasticity and cloaking. *Journal of the Mechanics and Physics of Solids*, 129:229–243, 2019.

[5] A.N. Norris and A.L. Shuvalov. Elastic cloaking theory. *Wave Motion*, 48(6):525–538, 2011. Special Issue on Cloaking of Wave Motion.